6 - (MathAA-W 2021-Paper 2/0-Q6) **-** *QUADRATICS*

[Maximum mark: 8]

(a) Prove the identity $(p+q)^3 - 3pq(p+q) \equiv p^3 + q^3$. [2]

The equation $2x^2 - 5x + 1 = 0$ has two real roots, α and β.

Consider the equation $x^2 + mx + n = 0$, where m, $n \in \mathbb{Z}$ and which has roots $\dfrac{1}{\alpha^3}$ and $\dfrac{1}{\beta^3}$.

(b) Without solving $2x^2 - 5x + 1 = 0$, determine the values of m and n. [6]

1 - (MAT-W 2013-Paper 2/0-Q3) **-** *EXPONENTIALS - LOGARITHMS, GRAPHS*

Consider $f(x) = \ln x - e^{\cos x}$, $0 < x \le 10$.

(a) Sketch the graph of $y = f(x)$, stating the coordinates of any maximum and minimum points and points of intersection with the x-axis. *[5]*

(b) Solve the inequality $\ln x \le e^{\cos x}$, $0 < x \le 10$. *[2]*

2 - (MAT-W 2013-Paper 2/0-Q13) **-** *FUNCTIONS - ROOTS, EXPONENTIALS - LOGARITHMS, INTEGRATION*

A function f is defined by $f(x)=\frac{1}{2}\left(e^{x}+e^{-x}\right)$, $x \in \mathbb{R}$.

(a) (i) Explain why the inverse function f^{-1} does not exist.

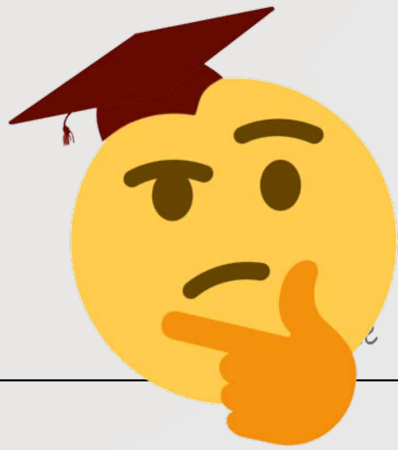
(ii) Show that the equation of the normal to the curve at the point P where $x=\ln 3$ is given by $9x+12y-9\ln 3-20=0$.

(iii) Find the x-coordinates of the points Q and R on the curve such that the tangents at Q and R pass through $(0, 0)$. *[14]*

(b) The domain of f is now restricted to $x \geq 0$.

(i) Find an expression for $f^{-1}(x)$.

(ii) Find the volume generated when the region bounded by the curve $y=f(x)$ and the lines $x=0$ and $y=5$ is rotated through an angle of 2π radians about the y-axis. *[8]*



3 - (MAT-S 2016-Paper 2/2-Q3) **-** *EXPONENTIALS - LOGARITHMS*

[Maximum mark: 6]

Solve the simultaneous equations

$$\ln \frac{y}{x} = 2$$

$$\ln x^2 + \ln y^3 = 7\,.$$

4 - (MAT-S 2016-Paper 2/1-Q7) **-** *EXPONENTIALS - LOGARITHMS*

It has been suggested that in rowing competitions the time, T seconds taken to complete a $2000\,\mathrm{m}$ race can be modelled by an equation of the form $T = aN^b$, where N is the number of rowers in the boat and a and b are constants for rowers of a similar standard.

To test this model the times for the finalists in all the $2000\,\mathrm{m}$ men's races at a recent Olympic games were recorded and the mean calculated.

The results are shown in the following table for $N = 1$ and $N = 2$.

N	T (seconds)
1	420.65
2	390.94

(a) Use these results to find estimates for the value of a and the value of b. Give your answers to five significant figures. [4]

(b) Use this model to estimate the mean time for the finalists in an Olympic race for boats with 8 rowers. Give your answer correct to two decimal places. [1]

It is now given that the mean time in the final for boats with 8 rowers was 342.08 seconds.

(c) Calculate the error in your estimate as a percentage of the actual value. [1]

(d) Comment on the likely validity of the model as N increases beyond 8. [2]

5 - (MAT-S 2017-Paper 2/2-Q6) **-** *EXPONENTIALS - LOGARITHMS*

[Maximum mark: 5]

Given that $\log_{10}\left(\frac{1}{2\sqrt{2}}(p+2q)\right)=\frac{1}{2}\left(\log_{10} p+\log_{10} q\right)$, $p>0$, $q>0$, find p in terms of q.

6 - (MAT-S 2019-Paper 2/1-Q11) **-** *EXPONENTIALS - LOGARITHMS*

Consider the equation $x^5 - 3x^4 + mx^3 + nx^2 + px + q = 0$, where $m, n, p, q \in \mathbb{R}$.
The equation has three distinct real roots which can be written as $\log_2 a$, $\log_2 b$ and $\log_2 c$.
The equation also has two imaginary roots, one of which is $d\mathrm{i}$ where $d \in \mathbb{R}$.

(a) Show that $abc = 8$. [5]

The values a, b, and c are consecutive terms in a geometric sequence.

(b) Show that one of the real roots is equal to 1. [3]

(c) Given that $q = 8d^2$, find the other two real roots. [9]

7 - (MathAA-S 2021-Paper 2/2-Q5) **-** *EXPONENTIALS - LOGARITHMS*

[Maximum mark: 7]

All living plants contain an isotope of carbon called carbon-14. When a plant dies, the isotope decays so that the amount of carbon-14 present in the remains of the plant decreases. The time since the death of a plant can be determined by measuring the amount of carbon-14 still present in the remains.

The amount, A, of carbon-14 present in a plant t years after its death can be modelled by $A = A_0 e^{-kt}$ where $t \geq 0$ and A_0, k are positive constants.

At the time of death, a plant is defined to have 100 units of carbon-14.

(a) Show that $A_0 = 100$. [1]

The time taken for half the original amount of carbon-14 to decay is known to be 5730 years.

(b) Show that $k = \dfrac{\ln 2}{5730}$. [3]

(c) Find, correct to the nearest 10 years, the time taken after the plant's death for 25% of the carbon-14 to decay. [3]